

Performance Testing

Date	15 March 2026
Team ID	TM-OPTICROP-04
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust, Postman
Type of Testing	Load Testing, Stress Testing
Target Module / API	ML Optimization Core Engine (`/api/v1/predict-yield` endpoint)
Test Environment	Cloud / Hosted Server (AWS EC2 t3.medium instance staging environment)
Test Date	12 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Baseline baseline simulation benchmarking standard dashboard traffic requests to the Telemetry API.	50	300	Response latency remains under 200ms with 0% error packet drops.
2	Peak harvest-season data sync load concurrent hits targeting the Crop Yield Prediction Engine.	200	600	Average response time stays below 500ms; HTTP 200 OK success codes returned.
3		500	120	System processes

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
	Stress testing rapid micro-climate sensor array batch submissions to check data pipeline queuing limits.			requests gracefully via message queue without dropping data entries.
4	Sudden weather alert notification spike stress test simulation hitting regional telemetry broadcast workers.	1000	60	Server utilization spikes up to 85% but successfully self-throttles without core service disruption.

Project Executable Files

Date	15 March 2026
Team ID	OC-2026-AGRO
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	Yes
5	Environment configuration file (.env.example or similar)	Yes
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	Yes
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes